

Bicol University Polangui
Bachelor of Science in Information Technology
Subject: IT 123 - Week 3

Lab Report 2

User & Group Management in Ubuntu Server

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1. Objectives

1. Create, modify, and delete user accounts on Windows and Linux.
2. Configure groups and assign users to groups.
3. Apply and test permissions on files and directories.
4. Troubleshoot common user-related issues.

2. Setup Summary

Item	Value	Notes
Users	faculty2, student4	Created with adduser
Groups	facultygrp, studentgrp	Created with groupadd
Group assignment	faculty2 → facultygrp student4 → studentgrp	usermod -aG
/facultydata	owner root:facultygrp, mode 770	Read/write for facultygrp only
/studentdata	owner root:studentgrp, mode 750	Read-only for studentgrp only

Figures 1–4 below show the corresponding terminal output for each setup step.

3. Commands Executed

3.1 Create users

`sudo adduser faculty2`

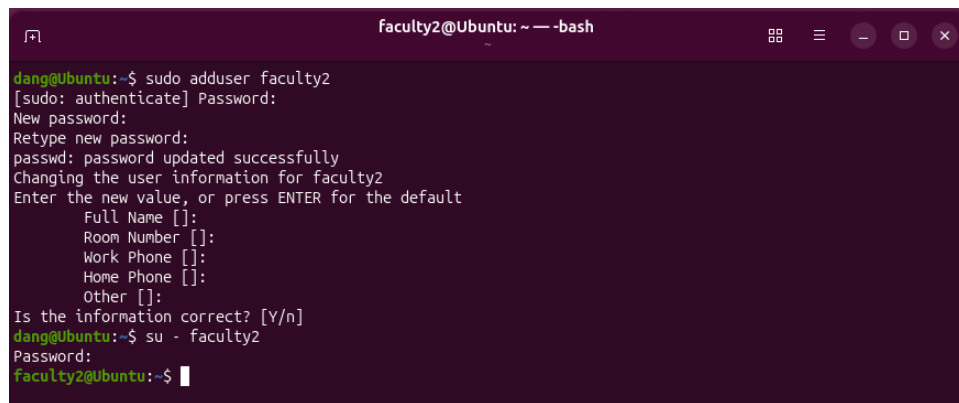
A terminal window titled 'faculty2@Ubuntu: ~ -- -bash' showing the process of creating a user. The user 'dang@Ubuntu' runs 'sudo adduser faculty2'. The system prompts for a password, then a new password, and then retype the new password. It then updates the password successfully and prompts for user information for 'faculty2'. The user enters 'Full Name []:', 'Room Number []:', 'Work Phone []:', 'Home Phone []:', and 'Other []:'. It then asks 'Is the information correct? [Y/n]'. The user runs 'su - faculty2' and enters the password. The prompt changes to 'faculty2@Ubuntu:~\$'.

Figure 1. Creating faculty2 and switching into the account.

```
sudo adduser student4
```

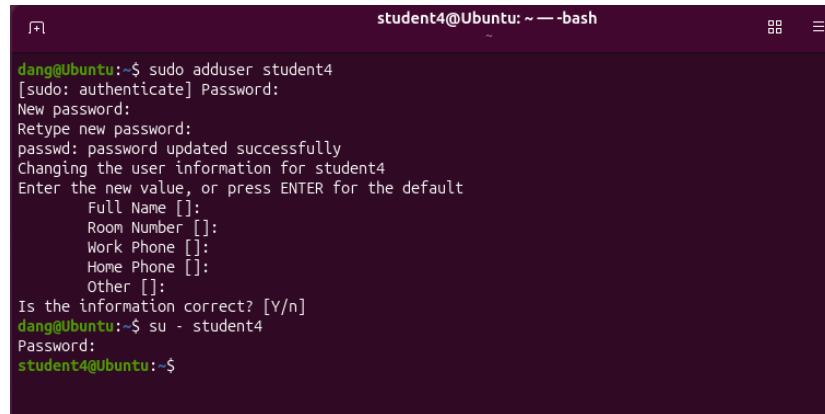
A terminal window titled 'student4@Ubuntu: ~ -- bash'. The user 'dang@Ubuntu' runs 'sudo adduser student4'. The system prompts for a password, then a new password, and then to retype the new password. After successful password entry, it prompts for user information: Full Name, Room Number, Work Phone, Home Phone, and Other. The user enters values for Room Number, Work Phone, and Home Phone. It then asks if the information is correct. Finally, the user runs 'su - student4' and is prompted for a password, then the prompt changes to 'student4@Ubuntu:~\$'.

Figure 2. Creating student4 and switching into the account.

3.2 Create groups and assign users

```
sudo groupadd facultygrp
```

```
sudo groupadd studentgrp
```

```
sudo usermod -aG facultygrp faculty2
```

```
sudo usermod -aG studentgrp student4
```

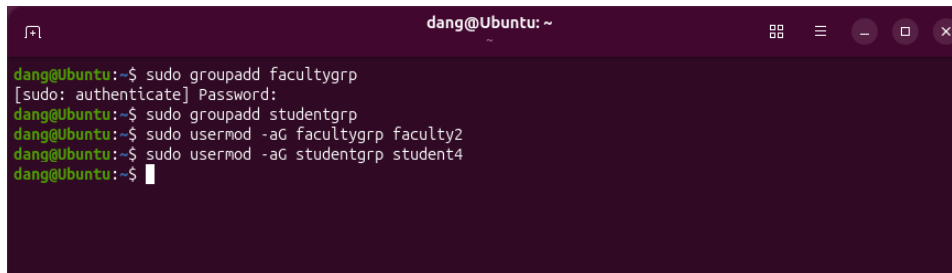
A terminal window titled 'dang@Ubuntu: ~'. The user 'dang@Ubuntu' runs four commands: 'sudo groupadd facultygrp', 'sudo groupadd studentgrp', 'sudo usermod -aG facultygrp faculty2', and 'sudo usermod -aG studentgrp student4'. Each command is preceded by a '[sudo: authenticate] Password:' prompt. The terminal shows the successful execution of each command.

Figure 3. Creating facultygrp and studentgrp, and assigning faculty2 and student4 to their groups.

3.3 Create directories and apply permissions

```
sudo mkdir /facultydata
```

```
sudo chown root:facultygrp /facultydata
```

```
sudo chmod 770 /facultydata
```

```
sudo mkdir /studentdata
```

```
sudo chown root:studentgrp /studentdata
```

```
sudo chmod 750 /studentdata
```

```
dang@Ubuntu: ~  
dang@Ubuntu:~$ sudo groupadd facultygrp  
[sudo: authenticate] Password:  
dang@Ubuntu:~$ sudo groupadd studentgrp  
dang@Ubuntu:~$ sudo usermod -aG facultygrp faculty2  
dang@Ubuntu:~$ sudo usermod -aG studentgrp student4  
dang@Ubuntu:~$ sudo mkdir /facultydata  
dang@Ubuntu:~$ sudo chown root:facultygrp /facultydata  
dang@Ubuntu:~$ sudo chmod 770 /facultydata  
dang@Ubuntu:~$ sudo mkdir /studentdata  
dang@Ubuntu:~$ sudo chown root:studentgrp /studentdata  
dang@Ubuntu:~$ sudo chmod 750 /studentdata  
dang@Ubuntu:~$
```

Figure 4. Full setup sequence: groups, group assignment, directories, ownership, and permissions.

4. Verification & Testing

User	Action	Directory	Expected	Actual Output
faculty2	touch (write)	/facultydata	Allowed	Succeeded, no error
faculty2	touch (write)	/studentdata	Denied	Permission denied
student4	ls (read)	/studentdata	Allowed	Listed contents, no error
student4	touch (write)	/studentdata	Denied	Permission denied
student4	cat (read)	/facultydata	Denied	Permission denied

Screenshot Evidence

Test 1–2: faculty2 write access

```
dang@Ubuntu:~$ su - faculty2  
Password:  
faculty2@Ubuntu:~$ touch /facultydata/test.txt  
faculty2@Ubuntu:~$ touch /studentdata/test.txt  
touch: setting times of '/studentdata/test.txt': Permission denied  
faculty2@Ubuntu:~$ exit  
logout
```

Figure 5. faculty2 succeeds writing to /facultydata, denied on /studentdata.

Test 3–4: student4 access within /studentdata

```
dang@Ubuntu:~$ su - student4  
Password:  
student4@Ubuntu:~$ ls /studentdata  
student4@Ubuntu:~$ touch /studentdata/test.txt  
touch: cannot touch '/studentdata/test.txt': Permission denied  
student4@Ubuntu:~$
```

Figure 6. student4 can list /studentdata (read) but is denied write access.

Test 5: student4 access to /facultydata

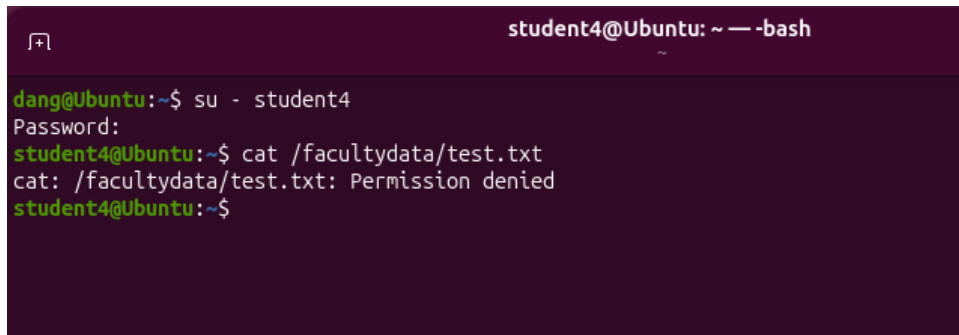
A terminal window with a dark purple background. The title bar at the top reads 'student4@Ubuntu: ~ — -bash'. The terminal shows a sequence of commands and outputs: 'dang@Ubuntu:~\$ su - student4' followed by 'Password:' (no input visible), then 'student4@Ubuntu:~\$ cat /facultydata/test.txt', which results in the error message 'cat: /facultydata/test.txt: Permission denied'. The prompt returns to 'student4@Ubuntu:~\$'.

Figure 7. student4 is denied read access to /facultydata.

5. Observations / Issues Encountered

- I first created the account as "Student2" with a capital S, but Linux treats usernames as case-sensitive, so that's actually a separate account from "student2." I had to delete it and create it again in lowercase to match what the exercise wanted.
- When I tested write access using a file that student2 already owned, the command went through even after I removed write permission. That's because touch on a file you already own just updates its timestamp, it doesn't need write access to the folder itself. Once I tested with a brand-new filename instead, the permission actually got enforced and the command failed like it should.
- After adding a user to a new group with usermod -aG, the change doesn't apply to a session that's already running. Switching to that user with su - starts a fresh session though, so the new group permissions worked right away without needing a full logout.

6. Conclusion

All five tests came back the way they were supposed to. faculty2 could read and write inside /facultydata but got blocked from /studentdata entirely, and student4 could only read inside /studentdata with no write access, and no access to /facultydata at all. That lines up with what the exercise was asking for, keeping faculty and student access completely separate on the Ubuntu server.